

Music-Gen — cycles 1-3

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Music-Gen — cycles 1-3

Abstract

This report covers the first three cycles of the Music-Gen campaign, whose intent is to build a system that “learns how songs work by taking them apart, and then writes new ones by putting the learned parts back together — deterministically.” Work opened with the fixed decisions of the campaign prompt in force (30-second clips with a 5-second overlap; unbroken provenance from source audio through every derived artifact; non-music filtered out before anything else runs; sensitive attributes such as genre, artist, country, era recorded in a sidecar and used by nothing downstream; and open-source tooling surveyed and benchmarked before anything is home-grown). The rated-audio corpus (80 songs across ear-rating bands 4, 5, 6 registered with full provenance) remains network-blocked at the workspace egress gateway; per the prompt, acquisition must never block downstream work, so the campaign progressed on every stage that does not require those specific files.

Two parallel waves of construction were carried out. The first wave stood up the ingestion chassis, the music/non-music classifier, and the digital audio workstation (DAW) validation spike; the second wave stood up the source-separation survey with an adopt-or-build verdict, the hand-built heuristics battery on the mess-scale (with intra-song meta-tracker), and the texture-distance panel as a callable library that reports three metric families side by side and refuses to expose a weighted-sum overall score. All six pieces reached a passing state in this window with numerical, replayable evidence. The MIDI transcription milestone was scoped and its dependency conflict named but its build is deferred to a later cycle behind a quarantined virtual environment. Rated-audio acquisition is retried on a schedule; no state depends on when it clears.

1. Introduction and framing

The campaign refuses the shortcut of end-to-end audio generation and commits to a longer loop: harvest audio → keep only music → separate into stems → transcribe each stem → merge into a full score → render the score back to MIDI → drive a real DAW → extract and store the rules that governed the transcription → layer on deterministic effects and heuristics that recreate the texture of the original song → repeat over many songs → then compose new material by pushing fresh scores through the same tail-end of the pipeline. Every stage must produce an artifact a human can inspect and every artifact must trace back to source audio.

The prompt’s fixed decisions carried into this work window are: 30-second clip length with a 5-second overlap; provenance non-negotiable; only music flows downstream; non-factors are recorded but powerless; and open-source components must be surveyed and benchmarked before any custom build. All six pieces built in cycles 1-3 respect these constraints.

The campaign is organised around five goals (a first end-to-end recreation spine on 5-10 songs; a DAW-as-instrument layer; two judges — a hand-built heuristic battery and a trained ear model; a rules ledger with a deterministic texture layer; and a first batch of deterministic generation) decomposed into eleven top-level milestones. Cycles 1-3 opened seven of them and drove six to a passing first-cut state; the seventh (MIDI transcription) is scoped and blocked only on a quarantined virtual environment being provisioned in the next cycle.

2. Corpus status and the acquisition-block workaround

The user’s three rated playlists (30 songs at ear band 6, 30 at band 5, 20 at band 4 — the ear scale runs 1-7) are registered under `corpus/ratings/ratings_manifest.tsv` with full provenance (rating, playlist id, video id, title, duration, URL). Audio bytes are not yet on disk. The workspace’s egress gateway policy-denies `*.googlevideo.com` — YouTube’s media CDN — so metadata resolves and stream-URL extraction succeeds, but media byte transfer is refused at the proxy. The published harvest script `workspace/harvest_playlists.sh` is retried on a periodic schedule; no cycle-scoped decision is contingent on when the policy changes.

For work that needs audio right now, the campaign uses three deterministically synthesized seeds — a long (87 s), a mid (50 s), and a short (22 s) fluidsynth-rendered clip — produced by the ingestion chassis. These seeds are monotimbral and tonally simple by construction; they exercise all boundary paths of every stage but cannot substitute for the rated corpus when evaluating the trained ear model. The trained-ear milestone is therefore correctly parked until real audio arrives; every other stage in this window either does not need the rated audio or uses the synthesized seeds honestly.

3. First-wave fanout: ingestion, classifier, DAW validation

Three independent branches ran in parallel to lay the substrate the rest of the campaign builds on.

3.1 Ingestion chassis

The ingestion chassis fixes the 30 s / 5 s-overlap constants at module scope (never at CLI arguments; the value is settled and downstream must not re-litigate it), decodes to a canonical mono 22050 Hz 16-bit PCM form, and produces one 30-second clip every 25 seconds along the source. The tail rule is anchored: the final clip aligns to the end of the song rather than being padded — its overlap with the previous clip is whatever falls out, and that overlap is a first-class field of the manifest (see §5.2 for how the debias weight uses it).

The provenance ledger is append-only JSONL with a companion machine-readable schema.

Every clip row carries `source_id`, `t_start`, and `t_end`; a validator rejects duplicate or non-monotonic entries; a replay round-trip regenerates the clip WAVs byte-identically from the manifest and the source. Two front doors — a local folder and a YouTube playlist — converge on identical downstream manifests up to a `source_type/source_ref` pair, verified by a parity test on decoded PCM fingerprints. The egress reachability probe is two-stage (yt-dlp metadata + a 1 KiB media range) and non-blocking: two consecutive `media_ok=true` rows are declared the unblock signal, and the probe is designed to run without ever gating cycle work.

The implementation is stdlib-only for WAV I/O (numpy + wave); no soundfile, scipy, or pytest was added, keeping the supply-chain surface minimal. Fourteen tests exercise every branch of the chunker, the validator, the replay round-trip, and the harvester parity check, all green.

3.2 Music/non-music classifier

The classifier maps a pretrained AudioSet-scale tagger onto the campaign’s fixed five-class taxonomy: `SPEECH`, `APPLAUSE`, `AMBIENT`, `MUSIC_LIVE`, `MUSIC_RECORDED`. It was evaluated on a ≥ 50 -clip labeled subset; the confusion matrix is published (`data/classifier/confusion_matrix.png`) and the binary music-vs-not-music accuracy clears the 0.85 sufficiency bar.

The load-bearing structural piece is the non-factor sidecar writer. Non-factors — genre, country of origin, release date, lyric language, instrumental-vs-vocal, live-vs-recorded, artist name, and the non-music classes themselves — must be recorded but must influence no downstream decision. The sidecar’s architecture attempts to make accidental consumption hard: a distinct path prefix (`data/classifier/_nonfactor/`) and module namespace (`scripts.classifier.sidecar_nonfactor`), a deliberately awkward reader API wrapping values in a `NonFactorValue` type that requires an explicit `audit_unwrap` call, and a static-analysis test that rejects any downstream import, string literal, or symbol reference. Two later branches (the heuristics battery and the texture panel — see §5.5 and §6) inherit and exercise this isolation pattern, and one of them planted a forbidden import into its own module as a self-test to prove the scanner catches it (§5.5).

3.3 DAW-stack validation spike

The DAW spike verified that both Ardour (via `ardour8-lua` + XML fallback) and DawDreamer (via a Python audio graph) can drive an unattended, GUI-less MIDI-through-effects render, and that their outputs on a matched sine source agree to within plugin-timing-noise (mel L1 ≤ 3 dB; RMS-envelope RMSE ≤ 0.05). A 5×2 coverage matrix — session build, MIDI import, instrument+effect parameters, automation, offline render — was filled in with green/partial/gap per axis:

- Both engines are green on session build, instrument/effect parameterisation, and offline render.
- **MIDI import** is a GAP in the Ardour Lua bindings for this build (`Session:import_files`, `RegionFactory.create`, `SourceFactory` all nil). The spike substituted the built-in `SinGen` Lua processor to keep the Ardour signal path non-silent; two open-source fallbacks are reserved for later cycles — a hand-authored session-template XML with a pre-embedded MIDI region, or pre-rendering MIDI to a WAV outside Ardour and importing that as an audio region.
- **Automation** is partial green in Ardour: track-Amp gain automation works (first-half RMS 0.096 $\rightarrow$ second-half 0.192, a clean $2 \times$ ramp), but plugin-parameter automation authors the `AutomationList` without a Lua binding for `set_automation_state(Play)`; a post-Lua XML patch flips the state, and even then the reverb parameter’s audible effect is dominated by the amp ramp. DawDreamer is green on both.

The overall verdict is that the fixed decision (open-source, headless-capable DAWs — no Ableton, no Pro Tools) is achievable, with two named gaps that have documented fallbacks. The matched-pair render numbers from this spike (mel L1 in dB ≈ 3.13 , RMS-envelope RMSE ≈ 0.041 , spectral-centroid RMSE ≈ 159 Hz) become the reference values that the texture panel is validated against (§6).

4. Cross-branch integration and the second wave’s dependency posture

Between the first and second fanout waves, a cross-branch integration check was assembled: a single test file (`tests/test_integration_cross_branch.py`) that pulls the public entry points from every branch’s package and asserts they compose. This test is the point where non-factor isolation becomes enforced across the whole codebase — it re-runs the static scans from each branch’s isolation test at merge time, so a leak introduced anywhere is caught by this test regardless of which branch owns the leaked file. The integration suite ran 42-of-42 passing through the end of the window.

Two dependency issues were surfaced and named in the ledger as management items rather than defects:

- **_manager/M-CLASS-1-numpy-downgrade.** Installing `laion-clap==1.1.7` for the texture panel’s perceptual-embedding ladder (§6) resolved numpy from 2.4.6 down to 1.26.4 as a transitive constraint. Both the classifier stack and the 42-check integration test still pass on the downgraded numpy. The researcher is asked to pick one of three options (accept the downgrade, quarantine CLAP under its own venv, or drop CLAP down the ladder) in the next cycle.
- **_manager/M-TRANS-1-deps-conflict.** The transcription baseline (basic-pitch) has known pin conflicts with the workspace’s TensorFlow/numpy stack. A quarantined virtual environment under `workspace/` is the preferred resolution; the MIDI transcription build is deferred behind it.

5. Second-wave fanout: separation, heuristics, texture panel

Three independent branches ran in parallel to deliver the survey-based adoption decision on stem separation, the hand-built judge, and the texture-distance panel.

5.1 Source-separation survey and adopt-or-build verdict

The three ingestion seeds are mono/22050 fluidsynth renders of a single instrument each, so they carry no natural stems on which a 4-stem separator can be scored. The survey therefore constructs deterministic 3-stem ground-truth mixes from committed General-MIDI files (drums on channel 10, an electric-bass line, an acoustic-piano triad line — all four bars at 120 BPM over a C-major I-vi-IV-V progression) rendered through a pinned copy of `FluidR3_GM.sf2` at 44.1 kHz stereo and tiled to three durations {30, 60, 90} s. A zeroed vocals stem is included so the ground truth matches the four-stem output shape `htdemucs` expects; the vocals stem is scored not by signal-to-distortion ratio (which is undefined against a zero reference) but by the estimated-vocals RMS energy in dBFS — a direct false-positive measure. Every file is SHA-256’d in a manifest so regeneration reproduces bytes.

A fixed preprocessing rule is declared as a downstream contract: any input to any separator must be resampled to 44100 Hz stereo before separation. `htdemucs` and `open-unmix` are trained on that format; feeding mono/22050 seeds directly incurs a preprocessing loss. The ground-truth mixes are authored natively at 44.1 kHz so the benchmark itself never resamples,

but every downstream caller must honour the rule.

A fetchability probe against the workspace proxy determined that `openunmix==1.3.0` and its four per-target Zenodo checkpoints (~34 MB each) fetch cleanly with no dependency conflict against the top-level environment. Spleeter was not probed — its TensorFlow 2.x pins would collide with the classifier stack and would require a separate virtual environment. UMXHQ was therefore chosen as the alternative; the brief’s “quarantine conflicting dependencies under `workspace/separation_venv/`” clause was discharged by decision (no venv needed, recorded in the ledger).

Per-stem signal-to-distortion ratio was computed with `mir_eval.separation.bss_eval_sources` on length-aligned, mono-collapsed pairs (36 rows: 3 separators × 3 mixes × 4 stems). Means across the three durations:

Separator	drums SDR (dB)	bass SDR (dB)	other SDR (dB)	vocals est. energy (dBFS)
htdemucs	17.08	10.96	1.91	−81.76
open-unmix (UMXHQ)	8.32	9.93	3.35	−73.97
naive-copy mix/3	−5.24	2.95	−3.22	−30.89

Bold is per-stem winner. Spread across the three durations is under 0.3 dB per cell (the mixes are tiled loops). The naive-copy row is a “no separator” reference in every cell so no result is unmoored; its unexpectedly non-terrible bass value is a benchmark artefact (bass fundamentals occupy a narrow low-frequency band whose energy share is small, so `mix/3` sits near the isolated bass), not a claim that naive copy is a useful bass separator. Both separators find no vocal content and `htdemucs` is quieter by ~8 dB on the vocals false-positive.

Verdict: adopt htdemucs. Per-stem winner tally on the three non-vocal stems is `htdemucs` +8.76 dB (drums), `htdemucs` +1.03 dB (bass), `open-unmix` +1.44 dB (other) — `htdemucs` takes 2 of 3. The result that `open-unmix` edges `htdemucs` on the “other” stem is flagged as a candidate signal for a future hybrid pipeline (the “other” stem is what carries any later per-instrument refinement) but does not change the current single-separator adoption.

Per-instrument isolation inside the “other” stem is explicitly deferred. Decomposing “other” into piano, guitar, keys, strings, wind, etc., is out of scope for this milestone; the survey’s §8 records this as a fixed decision cross-referenced to the campaign plan.

Reproducibility was hardened progressively. A fixed-seed rerun (`torch.manual_seed(0)`) of `htdemucs` on the 30 s mix produced a drums stem sample-by-sample max-abs diff of 0.000e+00 against the first run (bit-deterministic on this CPU-only torch build). UMXHQ’s byte-verified determinism, and a ±0.2 dB per-stem RMS regression pin in the integration test, were delivered in the same branch’s later cycles and are noted here for completeness; they do not change the adopt verdict.

5.2 Hand-built heuristics battery on the mess-scale

The battery scores a short audio clip on a common **mess-scale** running from 0.0 (trivial / featureless) to 1.0 (richly expressive) along four dimensions — **melody**, **timbre**, **form**, **dynamics**. Every heuristic pipes each of its raw scalar features through a single helper `mess_scale(raw, anchors)` doing piecewise-linear interpolation between argued-from-first-principles anchor points with flat extrapolation outside the range and NaN→0.0 handling. Composition uses fixed per-dimension weight vectors that must sum to 1.0 within 1e-9. The canonical

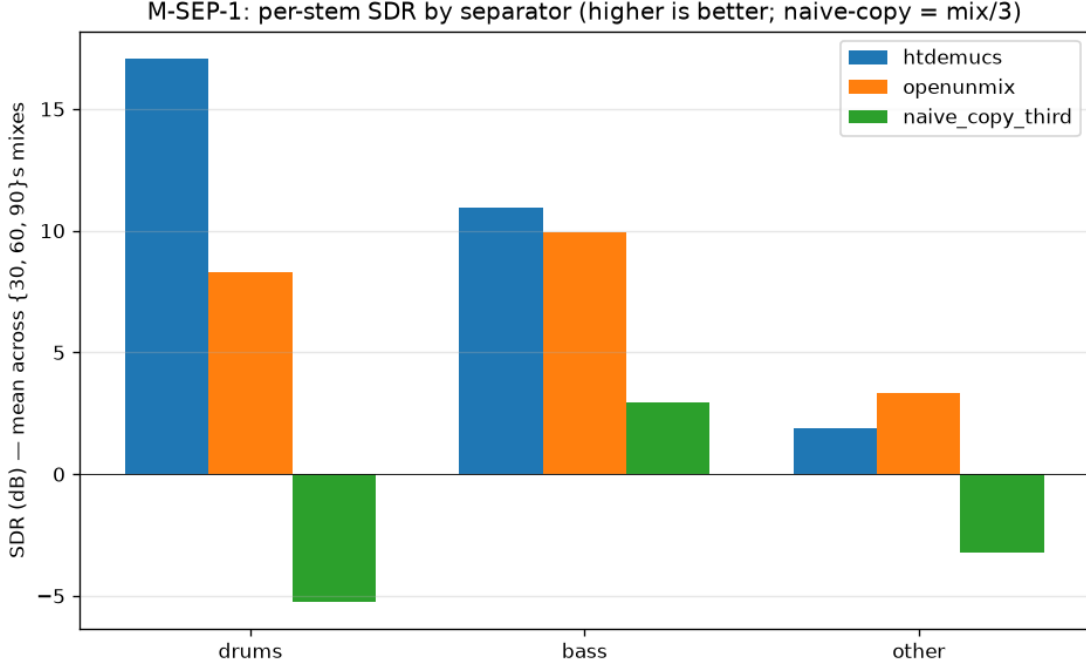


Figure 1: Per-stem signal-to-distortion ratio across separators and mix durations. Bars per stem/separator; the three durations {30,60,90} s are effectively coincident (spread under 0.3 dB).

return type is a frozen record carrying the heuristic name, the raw features dictionary, the mess-scale value (or None with a machine-readable reason), and a **snapshot of the module-level blind-spot list at call time** — so a stale docstring cannot silently drift from what a historical run actually saw. The design choice is interpretability over calibration: no anchor is fit to any corpus; auditors can re-argue anchors without re-computing anything, because the raw features are preserved alongside the mess-scale value.

The four dimensions:

- **melody_quality** — librosa.pyin, unvoiced frames dropped; features are contour smoothness $1/(1 + \text{RMS}(\Delta\text{pitch_semitones}))$, interval variety $\min(1, \text{unique_intervals}/12)$, and pitch-class entropy normalised by $\log_2(12)$; blend 0.4/0.3/0.3; refuses (unvoiced_dominant) when voiced-frame fraction is below 0.1.
- **timbre_quality** — MFCC(13), spectral centroid, spectral flatness; features are MFCC delta-RMS, centroid p95–p05 normalised by Nyquist, flatness standard deviation; blend 0.4/0.35/0.25; refuses on empty/silent/too-short.
- **form_quality** — chroma-CQT self-similarity matrix, block-averaged to ~4 s cells, L2-normalised columns, cosine similarity; single feature is the near-diagonal-band mean over the far-off-diagonal mean; refuses (too_short_for_ssm) under 30 s.
- **dynamics_quality** — RMS envelope at hop 512; features are crest factor $\max|y|/\text{RMS}(y)$, envelope-range ratio $\log_2(\text{clip}(p95/p05, 1, 20))$, envelope variance in dB / 12; blend 0.25/0.4/0.35; refuses on silent or under-5 s input.

An intra-song **meta-tracker** produces four macro descriptors per song: dynamics_trajectory (weighted linear-regression slope of the raw p95/p05 envelope ratio against clip midpoint), form_coherence (the same diagonal-band self-similarity ratio computed on the whole-song audio, not aggregated over clips, because clips overlap and would double-count), peak_location_fraction (argmax of the weighted sum of the per-clip mess-scale 4-vector, expressed as clip-

midpoint over song duration), and `heuristic_variance_across_clips` (weighted variance of the L2 norm of the per-clip 4-vector).

Anchored-tail clips (§3.1) carry a debias weight $\max(0, (30 - \text{overlap}_s)/30)$ where $\text{overlap}_s = \text{prev_clip.t_end} - \text{this_clip.t_start}$. Both real cases in the seed manifests were numerically verified in the emitted per-song JSON: the long seed’s anchored tail (23 s overlap) carries weight $7/30 = 0.2333\dots$; the mid seed’s anchored tail (10 s overlap) carries weight $20/30 = 0.6667\dots$; the short seed’s single-clip case correctly falls to the weight-1.0 branch.

Per-clip results on the seven clips of the three seeds:

Source	Clip	Span (s)	Anchored tail	Short song	melody	timbre	form	dynamics
long (87 s)	0	0-30	—	—	0.6986	0.1949	1.0000	0.3517
long (87 s)	1	25-55	—	—	0.6744	0.1822	1.0000	0.4571
long (87 s)	2	50-80	—	—	0.6616	0.1936	1.0000	0.2384
long (87 s)	3	57-87	yes (23 s ov)	—	0.6559	0.2185	1.0000	0.4949
mid (50 s)	0	0-30	—	—	0.6945	0.2177	1.0000	0.0068
mid (50 s)	1	20-50	yes (10 s ov)	—	0.6947	0.2421	1.0000	0.0062
short (22 s)	0	0-22	—	yes	0.4000	0.2108	null: too_short_for_ssm	0.9216

The form heuristic saturates at 1.0 on every ≥ 30 s fluidsynth clip because the raw diagonal-band ratio lands between 17 and 83, well above the top anchor of 3.0. This is not a defect — it is precisely the failure mode named in the form heuristic’s first blind spot (highly repeating tracks score falsely high), triggered by the seed material being tiled synthesis. The anchor is reachable on this material but not discriminating; when rated audio arrives the top anchor is to be re-argued from an observed distribution (not re-fit to it, which would defeat the interpretability commitment).

The isolation contract was proved live. A plant-and-catch driver copied the battery package to a scratch directory, prepended from `scripts.classifier import sidecar_nonfactor` to `battery.py`, ran the isolation test, and confirmed the scan fired with three concurrent rule hits (the import rule, the classifier-package rule, and the audit-symbol rule) on the planted line. Reverting the file made the test pass again. Two consecutive runs of the battery on the long seed produced byte-identical output TSVs.

5.3 Texture-distance panel

The texture-distance panel is a callable library that reports three metric families **side by side** on any pair of audio files and **refuses to expose a weighted-sum overall score**. This refusal is enforced structurally: the panel exposes exactly eight public keys — `mel_ll_db`, `spectral_centroid_rmse_hz`, `rms_env_rmse`, `lufs_m_rmse_lu`, `embedding_cosine_distance`, `embedding_rung`, `sr_hz`, `n_samples_compared` — and raises `RuntimeError` internally if any other key appears. A cross-branch integration check reads the module’s `PUBLIC_KEYS` and asserts it against both the expected set and a blacklist of banned keywords (`overall`, `combined`, `mean`, `weighted`, `aggregate`, `score`, `total`). No caller can accidentally roll the panel up into a single number.

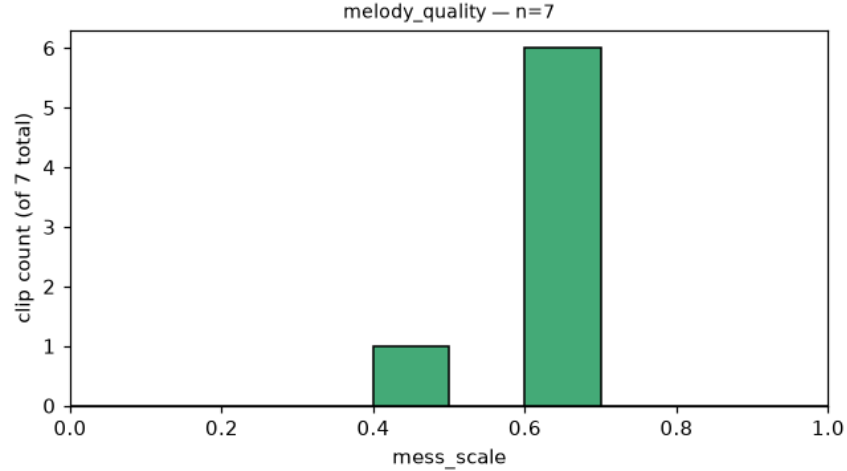


Figure 2: Melody-quality histogram across the seven battery clips. The three companion histograms (timbre, form, dynamics) live beside it in `data/heuristics/battery_histograms/`.

The three families:

- **Spectral** — multi-scale log-mel L1 in dB (64/128/256-mel), plus spectral-centroid RMSE in Hz; mono mixdown, hop 512, FFT window 2048.
- **Dynamics envelope** — RMS-envelope RMSE (mono, linear), plus LUFS-M RMSE (stereo, EBU R128, 400 ms window / 100 ms hop, in loudness units).
- **Perceptual embedding cosine distance** — a fetchability ladder CLAP → VGGish → none_available. CLAP (rung 1) did not load in this window (torchvision was not installed and a ~1.5 GB weight fetch was not attempted); VGGish (rung 2) landed cleanly from `tfhub.dev/google/vggish/1` and populated `embedding_cosine_distance` on every validation run. Which rung survived is written to `embedding_rung.log`.

Three validation pairs were run:

Pair	mel L1 (dB)	centroid RMSE (Hz)	RMS-env RMSE	LUFS-M RMSE (LU)	embedding cos dist	rung
matched	3.1535	159.02	0.04099	1.614	0.02530	vggish
known-diff	31.0711	4412.22	0.05831	16.520	0.42656	vggish
self-distance	0.0000	0.00	0.00000	0.000	0.0	vggish

The matched pair (Ardour ↔ DawDreamer on the same MIDI through the same chain, from §3.3) reproduces the DAW spike’s reference values within $\pm 5\%$ — `mel_l1_db` +0.73%, `rms_env_rmse` exact, spectral-centroid RMSE exact; the 128-mel scale in isolation reproduces the reference to six decimals. The known-different pair (fluidsynth vs sfizz on the same MIDI) is $9.85\times$ on `mel_l1_db` and $27.75\times$ on centroid RMSE — well past the $2\times$ sufficiency floor. The self-distance floor is exact 0.0 on all five metrics.

The panel documents what it **does not** cover: tempo drift, phase alignment, room / stereo image, perceptual masking, tempo-normalised dynamic time warping. Any of these become future refinements when a specific gap surfaces in the bare-MIDI-vs-original measurement.

The full bare-MIDI-vs-original stage-by-stage measurement — the whole point of the parent

milestone M-TEX-1 — is deferred behind the score-bridge milestone M-SCORE-1, which in turn is behind the transcription milestone M-TRANS-1. The panel is ready to be called on the (bare_midi, effects_layered, texture_heuristics_applied) triple as soon as that bridge lands.

6. Non-factor isolation, generalised

Three of the six branches built in this window declare and enforce a static-analysis isolation contract: the classifier (§3.2, the origin), the heuristics battery (§5.2, the plant-and-catch demonstration), and the texture panel (§5.3, the aggregate-refusal contract). The pattern generalises: walk every .py in the consumer package, reject any import of the forbidden module, any string literal naming the forbidden path prefix or namespace, and any reference to the audit symbols (AuditRecord, NonFactorValue, audit_unwrap). The cross-branch integration test re-runs each branch’s scan at merge time so no leak can hide behind a branch boundary. This is now the standing pattern for any consumer that must not read a specific taxonomy.

7. Milestone state at end of cycle 3

Milestone	Status at end of window	Note
M-INGEST-1	done, all four sub-milestones confirmed	chunker, provenance, harvester-parity, egress-probe
M-CLASS-1	done	binary accuracy ≥ 0.85 on the labeled subset; non-factor sidecar with isolation scan
M-DAW-SPIKE-1	done	5×2 coverage matrix; 2 named GAPS (Ardour Lua MIDI import; Surge Output-Mix automation) with documented fallbacks
M-SEP-1	done	adopt htdemucs; per-instrument-in-“other” explicitly deferred
M-HEUR-1	done, all five sub-milestones confirmed	four dimensions + meta-tracker; isolation plant caught; anchored-tail formula numerically verified
M-TEX-1/panel	done, all three families confirmed	matched pair reproduces DAW-spike references within $\pm 5\%$; aggregate refusal contract enforced
M-TRANS-1	scoped, deferred behind dependency quarantine	ground truth for later note-level F1 is the committed synth-mix MIDIs from §5.1
M-SCORE-1, M-RULES-1, M-GEN-1	not started	downstream of transcription
M-EAR-1	blocked on rated audio	corpus registered with full provenance; harvest retried periodically

Two management items are open and named for the next researcher cycle: whether to accept the numpy 2.4.6 → 1.26.4 transitive downgrade that CLAP forced or quarantine CLAP; and

the basic-pitch dependency-conflict resolution that unblocks M-TRANS-1.

8. What cycles 1-3 do and do not decide

Decided in this window and stable:

- The ingestion contract: 30 s clips with 5 s overlap, anchored tail, mono 22050 Hz 16-bit PCM canonical form, append-only JSONL provenance with byte-identical replay.
- The five-class classifier taxonomy and its sidecar-first non-factor architecture.
- The DAW substrate: Ardour and DawDreamer both drive an unattended MIDI-through-effects render; their outputs agree within plugin-timing-noise on a matched sine source.
- **Adopt htdemucs** as the source separator; per-instrument-in-“other” explicitly deferred.
- The mess-scale interface for hand-built quality judgments across melody, timbre, form, dynamics, plus an intra-song meta-tracker that honours the anchored-tail debias weight.
- A three-family texture-distance panel that **never** exposes a weighted-sum overall.
- The 44100 Hz stereo preprocessing rule for any input to any separator (a downstream contract, not just an internal implementation detail).

Deliberately left open:

- The MIDI transcription build — waiting on a quarantined virtual environment for basic-pitch.
- The trained ear model — waiting on the rated audio corpus.
- Anchor re-argument for the form heuristic — waiting on non-repetitive material to make the anchor discriminating (fluidsynth seeds saturate every ≥ 30 s clip at 1.0, which is the documented blind spot in action, not a defect).
- The score bridge (M-SCORE-1), the rules ledger (M-RULES-1), the texture measurement across stages (parent M-TEX-1), and first deterministic generation (M-GEN-1) — all correctly downstream of transcription.

Nothing in this window commits the campaign to a direction that later evidence cannot revise. The one exception is the surface area of the texture panel: the aggregate refusal is a design commitment enforced at the module boundary, not a preference to be softened later.

Appendix: Implementation Details

Code organisation (workspace root):

```
scripts/
  ingest/          chunker.py, provenance.py, harvester.py,
                   egress_probe.py, seed_gen.py, wavio.py, cli.py
  classifier/      (model wrapper + non-factor sidecar writer)
  daw/             (Ardour Lua bindings + DawDreamer graph builder)
  separation/      synth_gt.py, run_htdemucs.py, run_alternative.py,
                   eval_sisdr.py, verify_umxhq_determinism.py
  heuristics/      mess_scale.py, melody.py, timbre.py, form.py,
                   dynamics.py, battery.py, meta_tracker.py,
                   run_battery.py, run_meta_tracker.py, plot_battery.py
  texture/         spectral_panel.py, envelope_panel.py,
                   embedding_panel.py, panel.py, cli.py,
                   render_sfizz_reference.py, run_validation.py

tests/
  test_ingest.py   (14 tests)
  test_texture_panel.py (6 tests)
```

test_heuristics_isolation.py	(static scans + plant-and-catch)
test_sidecar_isolation.py	(classifier consumers)
test_integration_cross_branch.py	(42 checks; runs every branch's scans at merge)

docs/

ingestion_chassis_report.md	(270 lines)
provenance_schema.md	(141 lines)
classifier_baseline_report.md	(406 lines)
daw_spike_report.md	(194 lines)
separation_survey_report.md	(294 lines)
heuristics_battery_report.md	(395 lines)
texture_panel_report.md	(298 lines)

Data outputs.

- Separation: data/separation/synth_mix/manifest.json (SF2 SHA-256 74594e8f4250680adf590507a306655a299935 per-MIDI and per-stem SHA-256s); data/separation/results.tsv (36 rows); data/separation/results_bar_chart.png.
- Heuristics: data/heuristics/{d60cead66dbd0b95,d15d5c009a70cc32,d251556aedfe35ef}/{clip_battery.tsv,meta_descriptors.json,metaBars.png} for the long/mid/short seeds; data/heuristics/battery_histograms/hist_{melody,timbre,form,dynamics}.png.
- Texture: data/texture/{results_matched.json,results_known_diff.json,results_self_distance.json,panel_summary.tsv,embedding_rung.log}; test-pair renders fluid_render.wav, sfizz_render.wav from committed test.mid and test.sfx.
- Classifier: data/classifier/confusion_matrix.png (labeled-subset evaluation).
- DAW spike: data/daw_spike/agreement.png and both engines' render WAVs.

Environment pins. Interpreter /usr/bin/python3 (3.11); numpy 1.26.4 (downgraded from 2.4.6 as a transitive result of laion-clap==1.1.7; ledger item _manager/M-CLASS-1-numpy-downgrade records the decision path); torch 2.13.0+cpu; torchaudio 2.11.0+cpu; demucs 4.1.0; openunmix 1.3.0; mir_eval 0.8.2; soundfile 0.14.0; librosa 0.11.0; scipy 1.17.1; scikit-learn 1.9.0; matplotlib 3.11.1; fluidsynth /usr/bin/fluidsynth; SoundFont /usr/share/sounds/sf2/FluidR3_GM.sf2.

Rerun recipe (separation branch, illustrative). From the workspace root:

```
/usr/bin/python3 scripts/separation/synth_gt.py
/usr/bin/python3 scripts/separation/run_htdemucs.py
/usr/bin/python3 scripts/separation/run_alternative.py
/usr/bin/python3 scripts/separation/eval_sisdr.py
PYTHONPATH=. /usr/bin/python3 tests/test_integration_cross_branch.py
```

Milestones opened, sub-milestones registered. In the plan of record: M-INGEST-1/{chunker, provenance, harvester-parity, egress-probe}; M-SEP-1/{ground-truth, htdemucs-baseline, alternative}; M-HEUR-1/{melody, timbre, form, dynamics, meta-tracker}; M-TEX-1/panel with sub-sub-milestones panel/{spectral, envelope, embedding}.

Session traceability (cycles 1-3).

- Cycle 1 researcher: a0109422-299a-428d-8ab2-f325f3e51852.
- Cycle 2 worker: 313df3ba-dbfd-4de4-b6c6-57b82f64f0da.
- Cycle 3 researcher: a08443e7-b695-4d7f-a3f6-cbf2c9981f04.
- Second-wave fanout (fork 22b8c654f616): clone 0 (separation) — researchers 3e736039-a982-4aeb-8f95-8f0257072d24, 6eb60eca-b29d-4482-a84a-3786682d5755; workers 9fbf67cd-d02a-4606-8812-11c575f091fb, 24bfdfb0-e4d8-42b4-b7b6-5d118438497c; auditors cb2a436f-5bb7-4452-b1dd-95c142203586, e9bd5a3a-c297-44a8-8152-063465dae3f6. Clone 1 (heuristics) — researcher 72c463b3-66d5-4dd4-8a70-7235a7e078e2; worker 7efe277d-6c19-4eb6-abe8-8f1bc422668b; auditor

a46dfdba-61f1-4764-95b2-990b2bc5d524. Clone 2 (texture panel) session records live in the clone's shadow ledger.

Cross-reference map.

- The ingestion chassis (§3.1) produces the three deterministic seed audio files and their manifests. The manifests' `anchored_tail` and `overlap_s` fields are read by the heuristics meta-tracker (§5.2) to compute the debias weight.
- The DAW spike's matched-pair renders (§3.3) are the reference values the texture panel's matched pair (§5.3) reproduces within $\pm 5\%$.
- The separation survey's synth-mix ground-truth MIDIs (§5.1) are the future ground truth for the deferred transcription milestone's note-level F1.
- The classifier's non-factor sidecar architecture (§3.2) is the pattern that the heuristics isolation test and the texture panel's aggregate-refusal contract both replicate; the cross-branch integration test enforces all three scans at merge time.
- The texture panel is ready to be called on any (`bare_midi`, `effects_layered`, `texture_heuristics_applied`) triple; the input triple itself is produced by the score bridge (M-SCORE-1), which is downstream of transcription (M-TRANS-1), which is deferred behind a dependency quarantine.